Name: _____ Group: _____

Why H Is Not a Constant — Building the van Deemter Curve One Mechanism at a Time

Team roles for today (rotate from last session): Manager keeps the group on the current question — no running ahead to the simulator. Recorder writes the group's predictions and reconciliations on this sheet. Presenter reports at the checkpoints. Reflector notes where the group's thinking changed.

How this is graded: full credit requires, for every numbered question, your **written prediction before touching any control**, the **observed result**, and — where they disagree — a sentence on **what changed in your thinking**. A wrong prediction honestly reconciled earns full credit; a correct answer with no prediction does not.

One rule about the equations. The panel carries a collapsed **"The equations"** section under each plot. It is not hidden from you — open it whenever you want. But it only tells you what the formulas *are*. Every question below asks what they *do*, and that part is on you.

Where we left the Craig apparatus

In Exercise 1 you found that a band's variance grows in proportion to the distance it travels, and that the constant of proportionality — the plate height H — was just the stationary fraction q. It was a **single number**. Nowhere in it did the speed of the machine appear.

Q1 — No simulator. Two minutes, as a group. The Craig apparatus has no clock: each transfer is instantaneous, complete, and every tube reaches true equilibrium before the next transfer. List **three things a real packed column does that the Craig apparatus does not**, and for each one say whether you'd expect it to make bands wider or narrower.

STOP — one sentence from each group before you open the simulator.

Getting started

Open drylab → **Band shape** → **Mechanisms (van Deemter)** tab. Set the five sliders to:

Control	Setting
Particle size dp (µm)	3.0
Packing factor λ (eddy)	1.00
Diffusion Dm (×10^{-5} cm^2/s)	1.0
Cs — stationary-phase MT (ms)	2.00
Cm — mobile-phase MT (ms)	1.00

Leave **Log velocity axis** checked (it is on by default). Call this **Setting A**. Note the grey **stamp underneath the plot** — it repeats every slider value and adds two numbers the panel has worked out for you. You will use that stamp all period; you never need to export anything.

Q2 — By hand first. The eddy-diffusion term is A = 2·λ·dp. Compute it for Setting A, with units. Then answer: does your expression contain the velocity u anywhere, and what should that mean for the shape of the A trace on the plot?

One mechanism at a time

Q3 — Predict, then run. Set **both** Cs and Cm to 0. Before you touch anything: which curves will still be on the plot, and will the total still have a minimum? Sketch your prediction, then run it and read what the stamp says at the far right.

Follow-up — two real columns, one sentence each. (a) An open-tubular GC capillary has a stationary-phase film on its wall but **no packing**. Which term must be exactly zero for it, and why? (b) A capillary electrophoresis column has no packing **and no stationary phase at all**. Which terms survive there? (Your reading develops exactly this case.)

Q4 — The one that surprises people. Restore Cs = 2.00 and Cm = 1.00. Now predict *before moving anything*: if you make the particles bigger (dp 3.0 → 6.0), does the velocity at which H is smallest move left, move right, or stay put? Same question for packing the column better (λ 1.00 → 0.50). Commit to an answer, then run each and read u_opt off the stamp.

Change	Predicted u_opt	Observed u_opt	Observed H_min
dp 3.0 → 6.0			
λ 1.00 → 0.50			

(a) What does the A term control, and what does it *not* control? The formula that settles this has been in “The equations” the whole time — find it and say what feature of it explains your observation.

(b) Your reading (Wenzel) writes the eddy contribution as **A·v^(1/3)** and notes that the literature has never settled the question; this builder draws van Deemter’s original flat A. Suppose the v^(1/3) version is right. Would packing quality then move the optimum velocity? Which way? (Reason from the shape of the terms — no calculation needed.)

Q5 — Now move the B term. Return to Setting A, then raise Dm from 1.0 to 4.0 (a hotter, less viscous mobile phase). Predict the direction u_opt moves and by roughly what factor, then check the stamp.

	Predicted	Observed u_opt	Observed H_min
Dm 1.0 → 4.0			

Follow-up. You just quadrupled Dm, and only the B limb moved — yet “The equations” says Cm physically contains dp^2/Dm. The builder’s C sliders are deliberately lumped, so *you* have to supply the coupling: name one control in the **HPLC** section and one in the **GC** section that would move Cs or Cm in a real instrument, and identify the one control that moves **two van Deemter terms at once, in opposite directions**.

Q6 — Earn the number. “The equations” gives you u_opt = √(B/C) — so the formula is free. The work is the **units**: B = 2·γ·Dm with γ = 0.6, but Dm is displayed in units of 10^{-5} cm²/s, and the C sliders are in **ms**. Compute u_opt for Setting A in cm/s, showing the unit conversions, and compare with the stamp.

STOP — instructor checkpoint. Do not go on until your Q6 number matches the stamp.

Reading the curve like an operator

Q7 — Where the optimum actually lives. Set Cs = 3.00 and Cm = 0.00 (same total C as Setting A, all in one slider). Predict: at what velocity do the falling B/u limb and the rising Cs·u limb contribute **equally** to H — and where is that point relative to the diamond? Then hover both limbs to check, and record the value each contributes there.

Q8 — Missing the optimum. Back to Setting A. Predict which mistake costs more plate height: running at **half** u_opt, or at **double** u_opt? Commit, then hover the H total at both velocities (the stamp gives u_opt; double and halve it).

	u (cm/s)	H (µm)
u_opt ÷ 2		
u_opt × 2		

Reconcile what you find. Then the operator’s question: if the two mistakes cost the same H, why does every practical method — your reading gives GC numbers — deliberately run **above** the optimum rather than below it?

Q9 — Which column would you rather own? Set the sliders to **Setting B**: dp = 2.0, λ = 1.90, Cs = 0.50, Cm = 0.50. (Read that as a finer particle in a badly packed bed.) Record the stamp for both settings, then hover the bold **H total** curve at u = 0.5 cm/s in each.

	u_opt (cm/s)	H_min (µm)	H at u = 0.5 cm/s
Setting A (dp 3.0, λ 1.00)			
Setting B (dp 2.0, λ 1.90)			

First: what do you notice about the two H_min values? Second: if your job is to run this separation **as fast as possible**, which column do you want, and which single number in the table told you? Third: which slider group is responsible for that difference?

What H does to a peak you can see

Switch to the **Peak shape (overload)** tab. Set: Plates N = 100, Retention factor k = 2.0, Isotherm curvature = 0.00, Sample load = 1.0, Injection width = 1, Slow-site tailing τ = 0. Nothing is overloaded today — the isotherm is perfectly linear, so every effect you see below is broadening, not distortion.

A column of length L packed to plate height H has N = L/H plates. Making H smaller and making the column longer are two different routes to the same N.

Q10 — Predict, then run. You are about to change N from 100 to 400 — a fourfold gain, as if you had quartered H. Predict what happens to the peak’s **Sigma** and to its **relative** width (sigma divided by the first moment). Then read both off the readout table.

N	Predicted sigma	Observed Sigma	First moment	Sigma ÷ moment
100				
400				

Which of the two got better, and by exactly what factor? Where have you seen this factor before?

Q11 — Read a formula like a trend. “The equations” on this tab gives N_app = N·(1+k)/k. Back at N = 100: **before touching the k slider**, sketch N_app against k from 0 to 9. Mark on your sketch: what value does it approach at large k, and what happens as k → 0? Then verify at k = 0.5, 2.0 and 9.0.

k	Built-in N	Apparent N_app	First moment
0.5			
2.0			
9.0			

Now slide k all the way to 0 and write down exactly what the readout says. Explain *physically* why this model gives that answer for an unretained analyte — and what a real column would still do to that peak that this model leaves out.

Q12 — Nothing is overloaded, so why isn’t it symmetric? Return to k = 2.0. The curvature is still 0.00 and the load is still 1.0. Read **Asymmetry As (10 %)** and **Ideal-profile As (10 %)**. Then set N to 10, and to 800, reading As each time.

N	As (10 %)
10	
100	
800	

Explain the trend using the theorem you met in Exercise 1. What does this tell you about calling a peak “tailing” because As > 1?

Q13 — Broadening the column is not responsible for. Homework 1 asks what a **too-slow injection** does to a GC peak. Answer it with the simulator: at N = 100, k = 2.0, step **Injection width (plates)** from 1 to 20, watching both Sigma and the First moment.

Injection width	Sigma	First moment	N_app
1			
10			
20			

Two separate things happened. Name both, and say which one a chromatographer would call an extra-column effect. Check **Mass recovered** while you are here — what does it rule out as an explanation?

Going further — for the ambitious

A1 — Own the optimum. Starting from H = A + B/u + C·u, use calculus to derive u_opt and H_min. Then prove the result you discovered in Q4: show that ∂u_opt/∂A = 0 exactly. Finally, redo the derivative with Wenzel’s eddy term A·u^(1/3) in place of the flat A and show that your Q4(b) prediction about the direction of the shift was forced by the sign of one term.

A2 — Why UHPLC exists. Define the reduced plate height h = H/dp and reduced velocity ν = u·dp/Dm. Rewrite the A term in these variables and show it becomes a pure number,

independent of particle size. Given that a well-packed column reaches h_min ≈ 2 whatever its particle size, explain why H_min scales as dp while column backpressure scales as 1/dp², and what that trade buys.

A3 — Variances add, one more time (homework). In Q13 the injection spread the sample over the first w plates. Model this: a molecule entering at plate j only has (N − j) plates left to traverse, and j is uniform on 0…w−1. Using the law of total variance, derive an expression for the peak variance in terms of N, k and w. Predict Sigma at w = 10 and N = 100, k = 2, then check it against the readout. State which term is the column’s own broadening and which is the injector’s.